MACHINE LEARNING FOUNDATIONS

Problem Set 03
Data, Features, and Model Evaluation

From raw tables to robust models: mastering feature engineering, algorithm selection, decision boundaries, and performance metrics.

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Name: _____ **Date:** _____

Total Marks: 100 | **Suggested Time:** 75–100 minutes

The Big Idea

Raw data is messy, incomplete, and unstructured. Before an algorithm can learn, features must be cleaned, scaled, and encoded. Once trained, a model must be judged not just by blind accuracy, but by precision, recall, and generalization on unseen data.

Instructions
Answer every question. Show working for calculations and justify written answers. For multiple-choice questions, select one best answer. Marks are shown beside each question.

1 Part A: Data Representation and Feature Engineering

Big idea: Features shape how a model perceives reality. Poor scaling or bad encodings blind algorithms to patterns. *(25 marks)*

1. **Multiple choice [1 mark]:** A clinic keeps a table with these columns: `patient_id` (1043), `blood_type` (A, B, AB or O), `age` (34), `visit_date` (2025-03-02) and `doctor_notes` (free text). Which column is an unordered category?

   A. `patient_id`

   B. `blood_type`

   C. `age`

   D. `doctor_notes`

2. **Multiple choice [1 mark]:** A school table has a column called `education_level` with the values primary, secondary and university. How should this column be turned into numbers for a model?

   A. Give each value a random number

   B. Use 1, 2 and 3 in that order, because the order has meaning

   C. Delete the column because it is text

   D. Replace every value with 0

3. **Multiple choice [1 mark]:** A shop wants to use the column `shirt_colour` (red, blue, green) in a model. Why is it better to turn this into separate yes/no columns (one-hot encoding) than to use 1, 2 and 3?

   A. Because models cannot read the numbers 1, 2 and 3

   B. Because yes/no columns always make the model more accurate

   C. Because colours must be stored as pictures

   D. Because 1, 2 and 3 would wrongly suggest that green is bigger than red

4. **Multiple choice [1 mark]:** A table of 10,000 customers has an age column where 5% of the values are blank. What is a sensible first step?

   A. Fill the blanks with a typical value such as the median, and optionally add a column that says age was missing

   B. Delete the age column completely

   C. Fill every blank with 0

   D. Ignore the blanks and train the model anyway

5. **Multiple choice [1 mark]:** A model compares customers using age (from 20 to 70) and yearly income (from 10,000 to 200,000). It measures how close two customers are with straight-line distance. What goes wrong if the columns are not scaled?

   A. Age will dominate, because it has more digits

   B. Nothing, because distance treats all columns equally

   C. Income will dominate the distance, because its numbers are much larger

D. The model will refuse to run

6. **Multiple choice [1 mark]:** Two shoppers are described by how many of three items they bought. Shopper A bought $[2, 0, 3]$. Shopper B bought $[4, 0, 6]$: exactly twice as much of the same items. Which statement is true?

   A. The straight-line distance between A and B is 0

   B. Their cosine similarity is 1, because they point in the same direction

   C. They are not similar at all, because the numbers are different

   D. Their cosine similarity is 0

7. **Multiple choice [1 mark]:** You want to predict whether a customer will cancel a subscription. The table has these columns: `days_since_last_login`, `plan_type`, `customer_name`, and `cancellation_date` (it is filled in only after the customer cancels). Which two columns should you drop?

   A. `plan_type` and `days_since_last_login`

   B. `customer_name` only

   C. No column should be dropped

   D. `customer_name and cancellation_date`

8. **Multiple choice [1 mark]:** A weather table has `temperature_celsius` and `temperature_fahrenheit`, which always move together perfectly. What is the best action?

   A. Keep one of them, because the other adds no new information

   B. Keep both, because two columns are better than one

   C. Drop both

   D. Multiply them together

9. **Multiple choice [1 mark]:** A food-delivery company wants to predict delivery delays. Each order has a timestamp such as `2025-03-08 23:15`. Which new feature built from the timestamp is most likely to help?

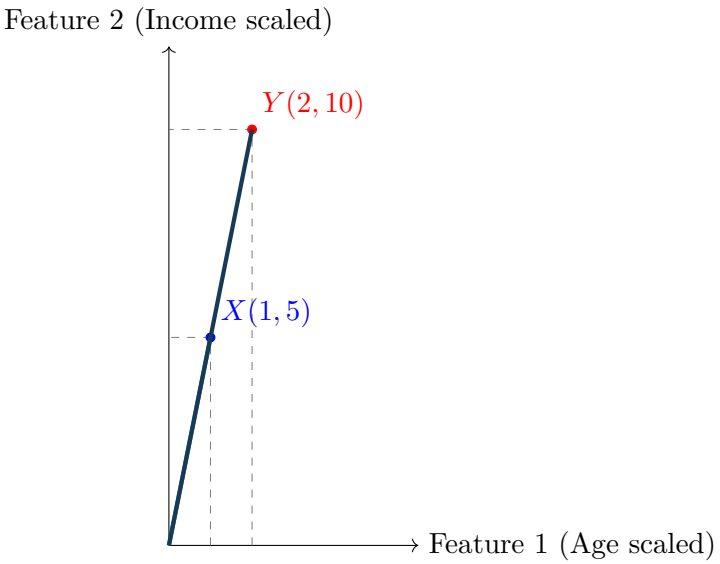
   A. The number of seconds since the year 1970 as a single big number

   B. Only the year

   C. The hour of the day and whether it is a weekend

   D. The first digit of the timestamp

10. **Graph Analysis [6 marks]:** Consider two customer vectors in a 2D feature space: Customer $X = (1, 5)$ and Customer $Y = (2, 10)$.



(a) Calculate the Euclidean distance between X and Y. *[3]*
(b) Explain why their cosine similarity is 1 despite having different absolute magnitudes. *[3]*

2 Part B: Choosing an Algorithm

Big idea: The structure of your problem dictates your model. Matching data constraints to algorithmic inductive bias is crucial. *(25 marks)*

11. **Multiple choice [1 mark]:** A shop has 50,000 customers and no labels. The marketing team wants to find natural groups of similar customers. Which type of learning fits best?

    A. Supervised classification

    B. Unsupervised learning (clustering)

    C. Supervised regression

    D. No learning is possible without labels

12. **Multiple choice [1 mark]:** A power company wants to predict tomorrow's electricity demand in megawatts using past demand and the weather forecast. What kind of problem is this?

    A. Classification, because the answer is a category

    B. Clustering, because there are many days

    C. Semi-supervised, because the future is unknown

    D. Regression, because the answer is a number

13. **Multiple choice [1 mark]:** A hospital has 100,000 X-ray images, but doctors have had time to label only 500 of them. Which approach best uses this situation?

    A. Semi-supervised learning: learn from the 500 labelled images and also from the many unlabelled ones

    B. Throw away the unlabelled images and use only 500

    C. Unsupervised clustering only, ignoring the labels

    D. Ask the model to guess without any data

14. **Multiple choice [1 mark]:** A bank must explain to every rejected applicant, in simple steps, why their loan was refused. Which model suits this requirement best?

    A. A very deep neural network

    B. k-means clustering

    C. A decision tree, because its yes/no questions can be read and explained

    D. A random forest of 500 trees

15. **Multiple choice [1 mark]:** A tool is trained to say "yes" or "no" to whether an email is spam, and it also reports a probability such as 0.92. Logistic regression is used for which job?

    A. Predicting a number such as a house price

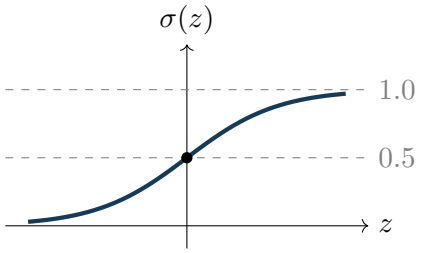
    B. Classification: choosing a category and giving a probability

    C. Grouping unlabelled data

    D. Compressing images

16. **Graph Analysis & Sigmoid Function [6 marks]:** Logistic regression outputs probabilities via the sigmoid function:

$$\sigma(z) = \frac{1}{1 + e^{-z}}$$



(a) If $z = 0$, what is $\sigma(z)$? *[2]*
(b) If a loan applicant's linear score is $z = 2.19$, compute their estimated probability of default and interpret what threshold ≥ 0.5 decides. *[4]*

17. **Short Structured Question [8 marks]:** Compare a single Decision Tree against a Random Forest. Why does a deep decision tree tend to overfit, and how does bagging in a Random Forest mitigate this variance? Give an intuitive explanation.

3 Part C: Judging a Model

Big idea: Accuracy is a dangerous illusion in imbalanced datasets. Precision and recall tell the true story of model failures. *(25 marks)*

18. **Multiple choice [1 mark]:** A bank checks 1,000 transactions, and only 1 is fraud. A lazy model says "not fraud" for every single transaction. What is the accuracy of this model, and what is the problem?

    A. Accuracy is 50%, which is as good as guessing

    B. Accuracy is 0%, so it is easy to spot as bad

    C. Accuracy is 99.9%, so it is an excellent model

    D. Accuracy is 99.9%, but it catches no fraud at all, so it is useless

19. **Multiple choice [1 mark]:** An airport scanner flags bags that might contain a dangerous item. Flagged bags get a manual check, which takes a couple of minutes. Which metric matters most?

    A. Recall: catch as many dangerous bags as possible, even if some flags are false alarms

    B. Precision: every flagged bag must be dangerous

    C. Accuracy only

    D. Neither matters

20. **Multiple choice [1 mark]:** An email service automatically moves emails to the spam folder. Users complain that an important job offer was lost there. Which metric should the team improve?

    A. Recall: catch every spam email at any cost

    B. The number of columns in the table

    C. Precision: when the filter says spam, it should be right

    D. The learning rate

21. **Multiple choice [1 mark]:** A spam filter flagged 100 emails as spam. Of these, 80 really were spam. In total the inbox held 200 real spam emails, so the filter missed 120 of them. What are its precision and recall?

    A. Precision 0.4 and recall 0.8

    B. Precision 0.8 and recall 0.4

    C. Precision 0.8 and recall 0.8

    D. Precision 0.4 and recall 0.4

22. **Multiple choice [1 mark]:** A model scores 100% on the data it was trained on, but only 60% on new data it has never seen. What is the most likely problem, and what could help?

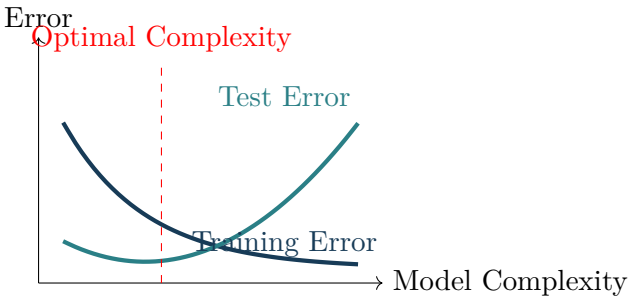
    A. Underfitting. Make the model even more complex

    B. The test data is too easy

    C. Nothing is wrong: training score is what counts

    D. Overfitting. Try a simpler model or more varied training data

23. **Graph Analysis: Overfitting Curves [8 marks]:**



(a) Explain why training error continues to drop as model complexity increases. *[4]*
(b) Identify the region where overfitting occurs and explain why test error spikes. *[4]*

4 Part D: Short Written Answers and Applications

Big idea: Synthesizing data pipeline design, metric evaluation, and rigorous written justification. *(25 marks)*

24. **Short answer [4 marks]:** You receive a table of past orders to predict which orders will be late. It has these columns: `order_id`, `customer_name`, `order_time` (a timestamp), `distance_km` (some blanks), `delivery_status`, and `refund_issued` (`yes`/`no`). In 3 to 5 sentences, say what you would fix, what you would drop, and one new feature you would add. Explain why.

> **Sample Response / Guidance:**
> First, fix the missing values in `distance_km` by imputing them with the median distance to ensure the model does not drop rows. Second, drop `customer_name` and `order_id` because they act as unique identifiers that cause overfitting rather than generalizable patterns. Third, drop `refund_issued` because a refund is only known after delivery completes, introducing data leakage. Finally, add a new feature such as `order_hour` extracted from the timestamp to capture peak traffic delays.

25. **Short answer [4 marks]:** A hospital wants a model that flags patients likely to be readmitted within 30 days, so that nurses can phone them. Calling a patient who did not need it is a small cost. Missing a patient who did need it can be serious. Say what kind of machine learning problem this is, which metric matters most, and why.

> **Sample Response / Guidance:**
> This is a supervised binary classification problem where the model predicts a categorical outcome (readmitted vs. not readmitted). High recall is the most critical metric because missing a high-risk patient leads to severe health consequences. Although false alarms result in unnecessary phone calls, this minor operational cost is far outweighed by the safety benefits of catching every true readmission risk.

26. **Comprehensive Confusion Matrix Analysis [9 marks]:** A disease diagnostic model evaluated on 1,000 patients yields the following confusion matrix:

	Predicted Positive	**Predicted Negative**
Actual Positive (Sick)	180 (TP)	20 (FN)
Actual Negative (Healthy)	50 (FP)	750 (TN)

(a) Calculate the model's overall Accuracy. *[3]*
(b) Calculate the model's Precision and Recall. *[4]*
(c) Explain why accuracy is misleading in this diagnostic context given the class imbalance. *[2]*